

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Poster Presentation

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Poster Presentation	Assessment:	Assessment Tool 2– Industrial Questions
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	P MOHAN BABU	Reg. No.:	192472254
Section / Batch:	CSE AI	Date:	24-08-2026

Code of Conduct

I P Mohan Babu / 192472254 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P Mohan Babu

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Marketing

Question: Before introducing a new advertisement, the conversion rate was 8%. After the campaign, 90 out of 1,000 visitors converted. Using a 95% confidence interval, determine whether the observed conversion rate provides evidence of improvement.

1. Identify the Population and Sample: Population: All visitors to the marketing campaign.

Sample: 1,000 visitors.

2. Identify the Parameter of Interest: Population conversion rate (p).

3. Calculate the Point Estimate: $\hat{p} = 90/1000 = 0.09$ (9%) **4. State the Null Hypothesis (H_0):** $H_0: p = 0.08$

5. State the Alternative Hypothesis (H_1): $H_1: p > 0.08$

6. Select the Significance Level (α): $\alpha = 0.05$

7. Calculate the Confidence Interval and/or Test Statistic: $SE =$

$$\sqrt{[(0.09 \times 0.91)/1000]} = 0.00905$$

$$ME = 1.96 \times 0.00905 = 0.0177$$

$$95\% \text{ CI} = 0.09 \pm 0.0177 = (0.0723, 0.1077)$$

$$95\% \text{ CI} = (7.23\%, 10.77\%)$$

8. Calculate the P-value: $z = (0.09 - 0.08)/\sqrt{[(0.08 \times 0.92)/1000]} = 1.17$
P-value = 0.122

9. Compare P-value with α : Since $0.122 > 0.05$, Do Not Reject H_0 .

10. Make an Industry-Oriented Conclusion: The campaign increased the conversion rate from 8% to 9%, but the improvement is not statistically significant. More sample data is needed.

2. Education Technology

Question: An EdTech company claims that its adaptive-learning platform improves the average test score to at least 75. A sample of 64 students has an average score of 78 with a standard deviation of 10. Use a 95% confidence interval to assess the claim.

1. Identify the Population and Sample: Population: All students using the adaptive-learning platform.

Sample: 64 students.

2. Identify the Parameter of Interest: Population mean test score (μ).

3. Calculate the Point Estimate: $\bar{x} = 78$

4. State the Null Hypothesis (H_0): $H_0: \mu = 75$

5. State the Alternative Hypothesis (H_1): $H_1: \mu > 75$

6. Select the Significance Level (α): $\alpha = 0.05$

7. Calculate the Confidence Interval and/or Test Statistic: $SE = 10 / \sqrt{64} = 1.25$

$ME = 1.96 \times 1.25 = 2.45$

$95\% \text{ CI} = 78 \pm 2.45 = (75.55, 80.45)$

8. Calculate the P-value: $z = (78 - 75) / 1.25 = 2.40$

$P\text{-value} = 0.0082$

9. Compare P-value with α : Since $0.0082 < 0.05$, Reject H_0 .

10. Make an Industry-Oriented Conclusion: The adaptive-learning platform significantly improves student performance, supporting the company's claim.

3. Automobile Industry

Question: A car manufacturer claims that its new model provides an average mileage of 20 km/litre. A sample of 50 cars has a mean mileage of 19.2 km/litre and a standard deviation of 2.5. Perform a hypothesis

test using the p-value approach.

1. Identify the Population and Sample: Population: All cars of the new model.

Sample: 50 cars.

2. Identify the Parameter of Interest: Population mean mileage (μ).

3. Calculate the Point Estimate: $\bar{x} = 19.2$ km/litre

4. State the Null Hypothesis (H_0): $H_0: \mu = 20$

5. State the Alternative Hypothesis (H_1): $H_1: \mu \neq 20$

6. Select the Significance Level (α): $\alpha = 0.05$

7. Calculate the Confidence Interval and/or Test Statistic: $SE = 2.5 / \sqrt{50} = 0.354$

$z = (19.2 - 20) / 0.354 = -2.26$

8. Calculate the P-value: P-value = 0.0236

9. Compare P-value with α : Since $0.0236 < 0.05$, Reject H_0 .

10. Make an Industry-Oriented Conclusion: The sample shows the average mileage is significantly lower than 20 km/litre. The manufacturer should review its claim.

4. Cloud Computing

Question: A cloud provider claims that its average server response time is 100 ms. A sample of 81 requests has a mean of 104 ms and a standard deviation of 12 ms. Test the claim using the p-value method.

1. Identify the Population and Sample: Population: All server requests.

Sample: 81 requests.

2. Identify the Parameter of Interest: Population mean response time

(μ).

3. Calculate the Point Estimate: $\bar{x} = 104$ ms

4. State the Null Hypothesis (H_0): $H_0: \mu = 100$ ms

5. State the Alternative Hypothesis (H_1): $H_1: \mu \neq 100$ ms

6. Select the Significance Level (α): $\alpha = 0.05$

7. Calculate the Confidence Interval and/or Test Statistic: $SE = 12 / \sqrt{81} = 1.333$

$z = (104 - 100) / 1.333 = 3.00$

8. Calculate the P-value: P-value = 0.0027

9. Compare P-value with α : Since $0.0027 < 0.05$, Reject H_0 .

10. Make an Industry-Oriented Conclusion: The average server response time is significantly higher than 100 ms. Server optimization is recommended.

5. E-commerce

Question: An online company claims that its product return rate is 10%. In a sample of 1,000 orders, 130 products were returned. Test whether the actual return rate is significantly higher than 10%.

1. Identify the Population and Sample: Population: All customer orders.

Sample: 1,000 orders.

2. Identify the Parameter of Interest: Population return rate (p).

3. Calculate the Point Estimate: $\hat{p} = 130 / 1000 = 0.13$ (13%) **4.**

State the Null Hypothesis (H_0): $H_0: p = 0.10$

5. State the Alternative Hypothesis (H_1): $H_1: p > 0.10$

6. Select the Significance Level (α): $\alpha = 0.05$

7. Calculate the Confidence Interval and/or Test Statistic: $SE =$

$$\sqrt{[(0.10 \times 0.90)/1000]} = 0.00949$$

$$z = (0.13 - 0.10)/0.00949 = 3.16$$

8. Calculate the P-value: $P\text{-value} = 0.0008$

9. Compare P-value with α : Since $0.0008 < 0.05$, Reject H_0 .

10. Make an Industry-Oriented Conclusion: The return rate is significantly higher than 10%. The company should improve product quality and customer satisfaction.